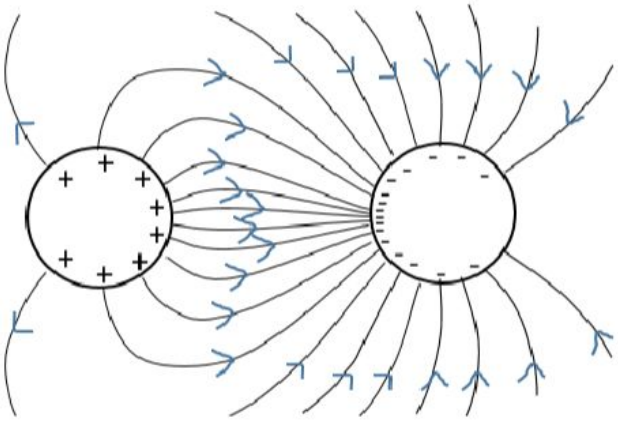
Answer **all** the questions in this Section in the spaces provided.

1	**(a)**	A positively charged metal sphere A is brought close to a negatively charged metal sphere B as shown in Figure 1.1. The charge on metal sphere B is twice that of the charge on metal sphere A. On Figure 1.1, draw lines to represent the electric field around the spheres.	
			
		Figure 1.1	[2]
	L2	Solution:  Marks scheme: • 1 mark: **Field pattern** (correct shape & variation of density of field lines), **AND**, **Direction** of field lines (arrows point away from A and towards B). **B1** • 1 mark: Details: **number** of field lines around B **twice** that around A (**at least 6 field lines around A and 12 field lines around B to sufficiently show trend/pattern**), **AND**, field lines drawn **perpendicular to the surface** of each **metal** sphere. **B1** OR • 1 mark: **Field pattern:** correct shape & variation of density of field lines, **number** of field lines around B **twice** that around A (**at least 6 field lines around A and 12 field lines around B to sufficiently show trend/pattern**). [B1] • 1 mark: **Direction** of field lines, including field lines drawn **perpendicular to the surface** of each **metal** sphere. [B1]	

(b) Point P is at a distance x from the centre of sphere A along the line joining the centres of the two spheres as shown in Figure 1.2. The radius of A and B is 15 mm and the distance between the centres of the spheres is 80 mm.

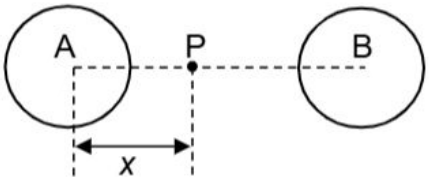


Figure 1.2

The variation with x of the electric potential V at P is shown in Figure 1.3.

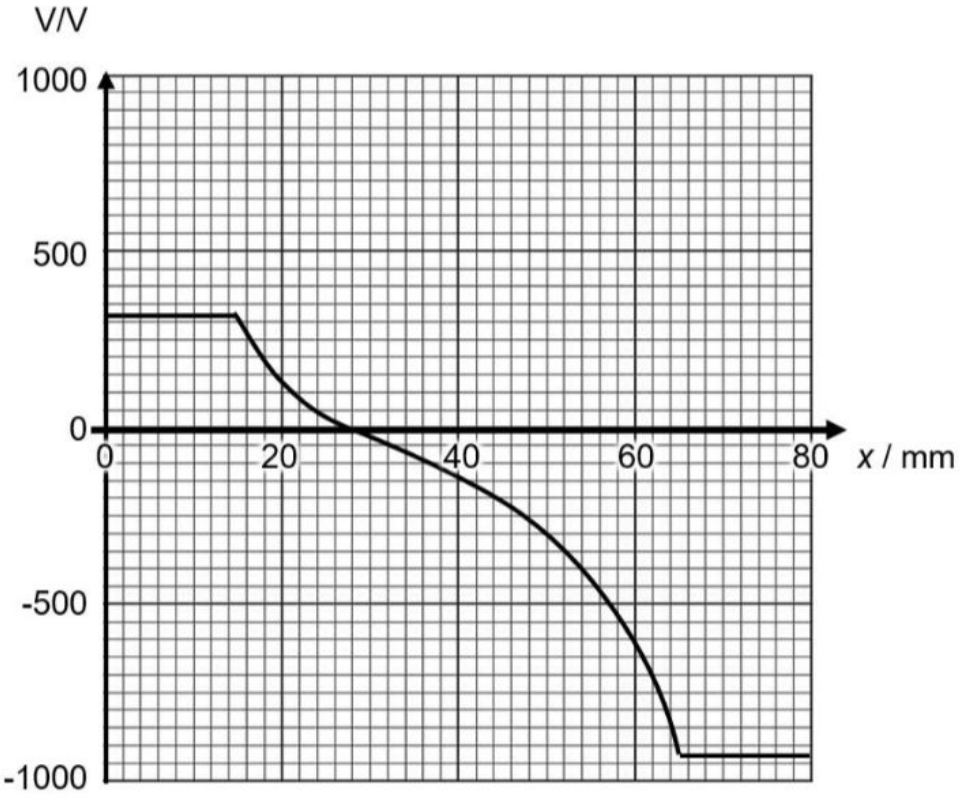


Figure 1.3

(i) Determine the magnitude of the electric field strength at P where x = 40 mm.

electric field strength = ... V m^{-1} [2]

L1
- **Tangent at x = 40 mm** must be **drawn long enough**
- **2 coordinates** along the tangent must be **far apart** enough

M1

$|E| = \left|\frac{dV}{dx}\right|$

= magnitude of gradient of V-x graph at x = 40 mm

= 350 – (-600) / 80 x 10^{-3}

= 1.19 x 10^{4} V m^{-1}

- **Accept answers that show correct read-offs, accurate to half-smallest division, of the 2 coordinates.** **A1**

(ii) An electron is released from rest at point P where x = 40 mm.

1. Describe and explain the motion of the electron as it travels 20 mm along the line joining the centres of the spheres.

.....

.....

.....

.....

.....

.....

..... [3]

L2

$a = \frac{F_{net}}{m} = \frac{F_E}{m} = \frac{eE}{m}$

where $E = -\frac{dV}{dx}$

Gradient of the V-x graph is proportional to the electric field strength which is proportional to acceleration. **B1**

When gradient decreases from 40 mm to 28 mm, acceleration decreases. **B1**

➔ **The electron moves to the left towards A, speeds up at a decreasing rate.**

When gradient increases from 28 mm to 20 mm, acceleration increases. **B1**

➔ **The electron continues moving to the left towards A, speeds up at an increasing rate.**

Note that when the electron travels 20 mm from x = 40 mm, it will not yet reach sphere A.

2. Determine the speed of the electron when it has travelled 20 mm along the line joining the centres of the spheres.

				speed = ……m s^{-1}	[4]
		L2		Moving 20 mm towards A, electron moves from x = 40 mm to x = 20 mm **increase** in potential, **ΔV = 125 – (-150) = 275 V** *[Also accept V_{40mm} between -125 V to -150 V; V_{20mm} between 125 V to 150 V]* Since the **electron is negatively charge**, it experiences a decrease in electric potential energy when potential increases. **decrease in** potential **energy**, **ΔU** = **$\lvert q.\Delta V \rvert$** = **$1.60 \times 10^{-19} \times 275 = 4.40 \times 10^{-17}$ J** **By conservation of energy,** increase in KE = decrease in electric potential energy final KE – initial KE = ΔU $\frac{1}{2} m v^2 - 0 = 4.40 \times 10^{-17}$ **$\frac{1}{2} (9.11 \times 10^{-31}) v^2 - 0 = 4.40 \times 10^{-17}$** **$v = 9.83 \times 10^{6}$ m s^{-1}** ***Allow partial ECF (award C1 & 1st M1 marks) if candidate in part 1 wrongly deduced that the electron moved to the right from x = 40 mm to 60 mm instead. No mark for 2nd M1 and A1 as it propagates the incorrect energy transformation and so wrong physics.***	**C1** **M1** **M1** **A1**

[Total: 11]